GREEN TEA LCA

Forest Residue Bioenergy & SAF Techno-Economic Analysis

Scenario	Biofuel Scenario
Mode	SAF
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NPV (Equity)	$-201.23M	**TCI**	$1106.53M
Equity IRR	5.50%	**SAF Output (Yr1)**	25.86M gal/yr
MFSP — SAF	$5.4780/gal	**Total Biomass**	0.0k odt/yr

1. Supply Chain & Transport

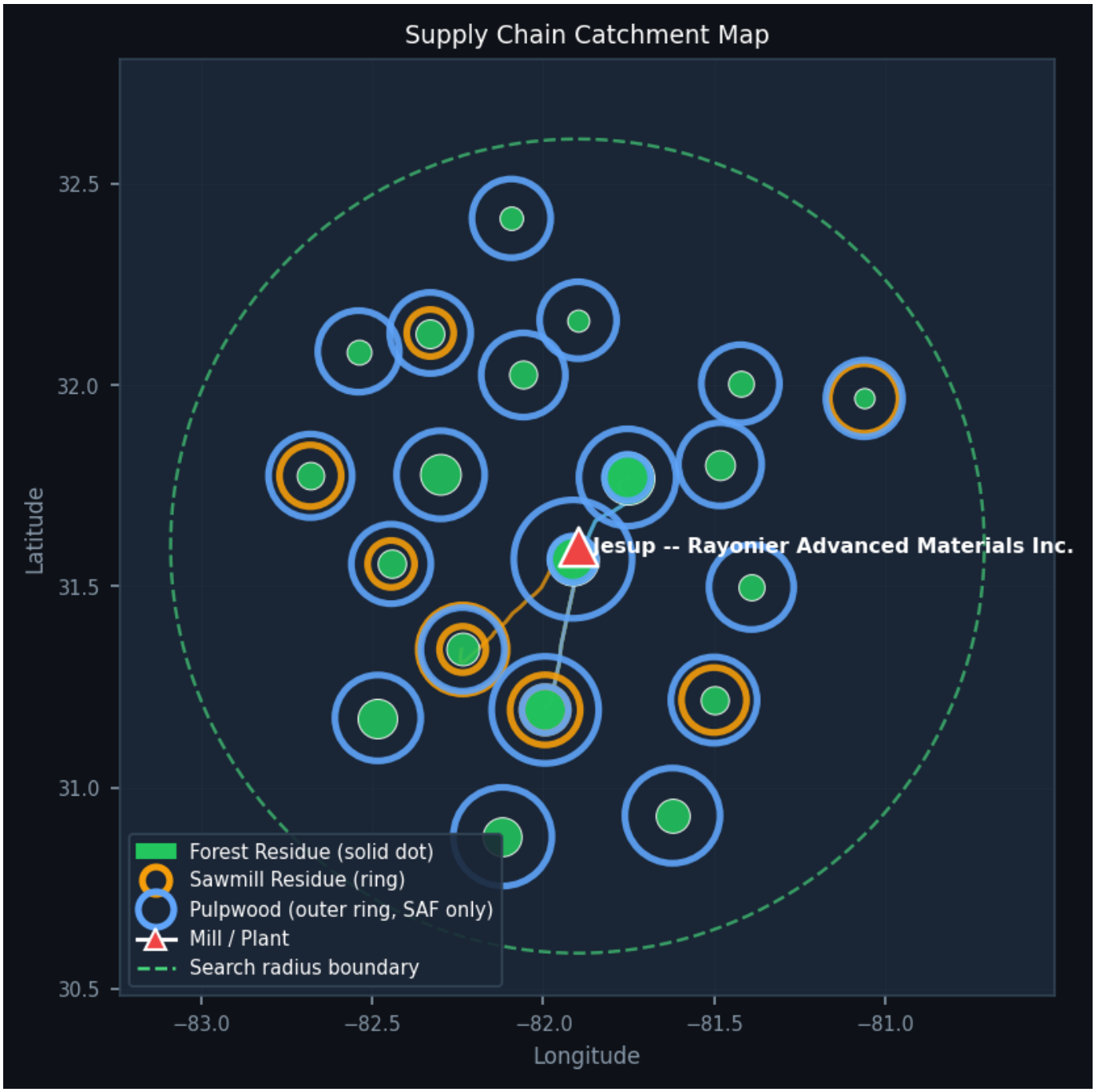
Source	Counties	Base Supply (k odt/yr)	Obtainability	Effective (k odt/yr)	Haul Dist (mi)	Transport Opt.	Delivered ($/ODT)
Forest Residue	3	143.2k	80%	79.3k	16	1.1	$63.20
Sawmill Residue	2	386.0k	90%	347.4k	32	3.1	$25.24
Pulpwood	3	432.3k	90%	389.1k	14	4.1	$27.23

Mill / Plant	Jesup -- Rayonier Advanced Materials Inc.
Search Radius	70 mi
Coordinates	31.6000°N, -81.9000°W

Transport Inputs

Parameter	Value
Avg. Truck Speed	20 mph
Forest Transport Option	-
Sawmill Transport Option	-
Pulpwood Transport Option	-

Forest Obtainability	100%
Sawmill Obtainability	100%
Pulpwood Obtainability	100%



Supply catchment map — Jesup -- Rayonier Advanced Materials Inc., 70-mile radius. Green: forest residue, amber: sawmill residue, blue: pulpwood. Dot size ∝ supply volume. Dashed circle = search boundary.

2. Economic Performance

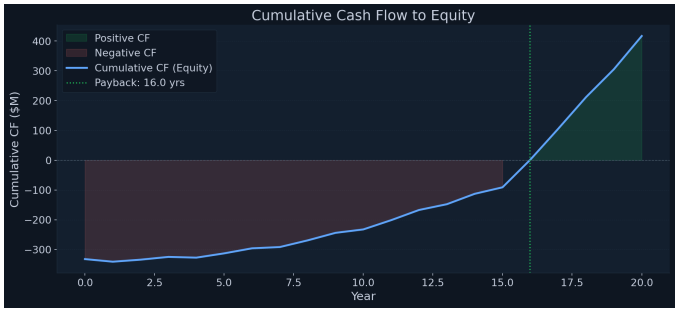
Inputs

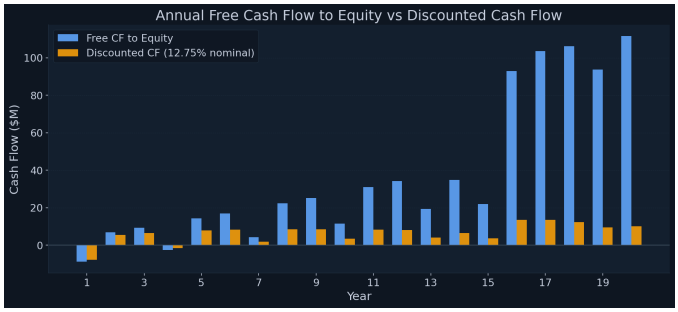
Analysis Year	2025	**Discount Rate**	10.0%
Distillate	distillate 1	**Inflation**	2.5%
SAF Price	$6.09/gal	**Debt Fraction**	70%
Diesel Price	$3.90/gal	**Loan Rate**	8.0%
Naphtha Price	$2.84/gal	**Loan Term**	15 yr
Plant Life	20 yr	**Federal Tax**	21%
CPI	321.05	**State Tax**	7%
Degradation	0.0%/yr	**Price Escalation**	2.5%/yr
		Fuel Escalation	2.5%/yr
		Cost Escalation	2.5%/yr

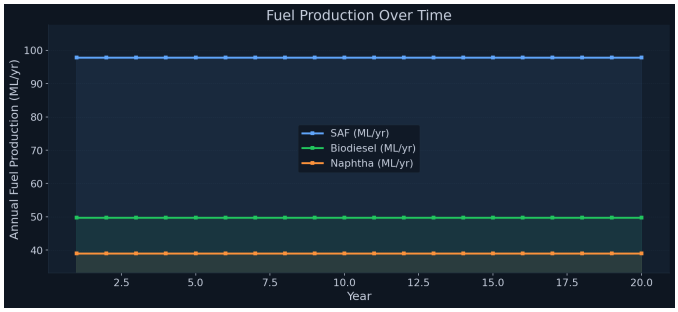
Outputs

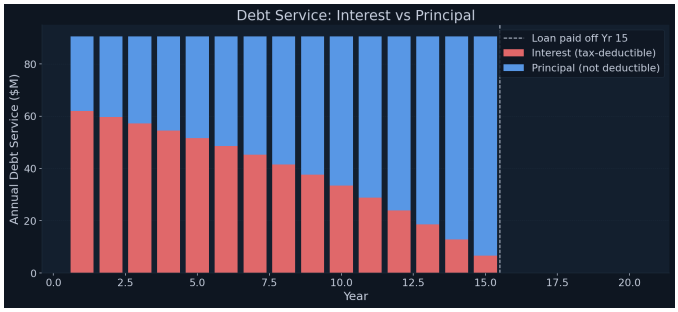
Metric	Value	Notes
TCI	$1106.53M	Total Capital Investment
FCI	$962.20M	Fixed Capital Investment
Year 1 Revenue	$238.00M/yr	
NPV (Equity)	$-201.23M	Negative = below hurdle rate
Equity IRR	5.50%	
Payback Period	16.0 yr	
MFSP — SAF	$5.4780/gal	At NPV=0
MFSP — Diesel	$5.5686/gal	At NPV=0
MFSP — Naphtha	$5.0816/gal	At NPV=0
SAF Production	25.86M gal/yr	Year 1
Diesel Production	13.15M gal/yr	Year 1
Naphtha Production	10.30M gal/yr	Year 1

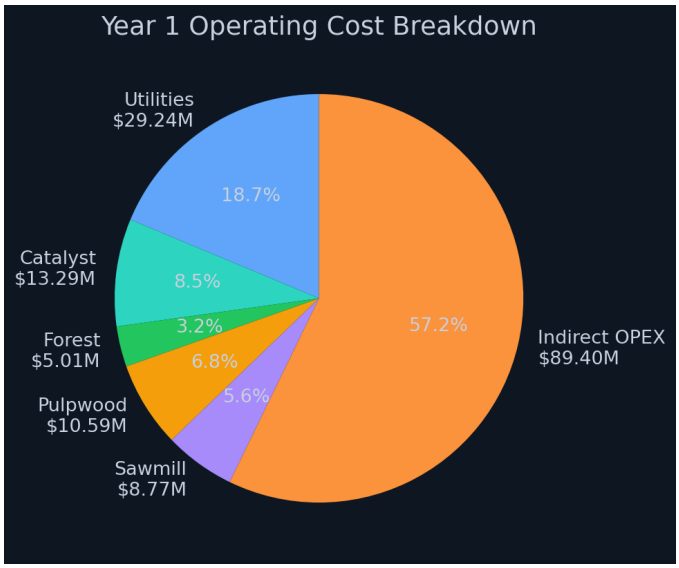
Cash Flow Plots






Fuel Production Over Time
Annual Fuel Production (ML/yr)
100
90
80
70
60
50
40
SAF (ML/yr)
Biodiesel (ML/yr)
Naphtha (ML/yr)
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Year


Debt Service: Interest vs Principal
Annual Debt Service ($M)
80
60
40
20
0
Loan paid off Yr 15
Interest (tax-deductible)
Principal (not deductible)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Year


Year 1 Operating Cost Breakdown
Utilities
$29.24M
18.7%
Catalyst
$13.29M
8.5%
Forest
$5.01M
3.2%
Pulpwood
$10.59M
6.8%
Sawmill
$8.77M
5.6%
Indirect OPEX
$89.40M
57.2%

3. Lifecycle GHG Emissions

Emission Factor Inputs

Parameter	Value	Notes
Moisture Content	40%	Feedstock
Truck Fuel (L/t-mi)	0.000094	Transport EF
Transport CO_2 EF	2.680 kg/L	Diesel combustion
Transport CH_4 EF	2.7186×10^{-5} kg/L	
Transport N_2O EF	3.6488×10^{-6} kg/L	
Processing CO_2 EF	2.680 kg/L	Diesel
Processing NCV	35.8 MJ/L	Diesel
Processing CH_4 EF	0.100 g/MJ	
Processing N_2O EF	0.100 g/MJ	

LCA not run for this scenario.

4. Avoided Emissions Analysis

LCA not run for this scenario.

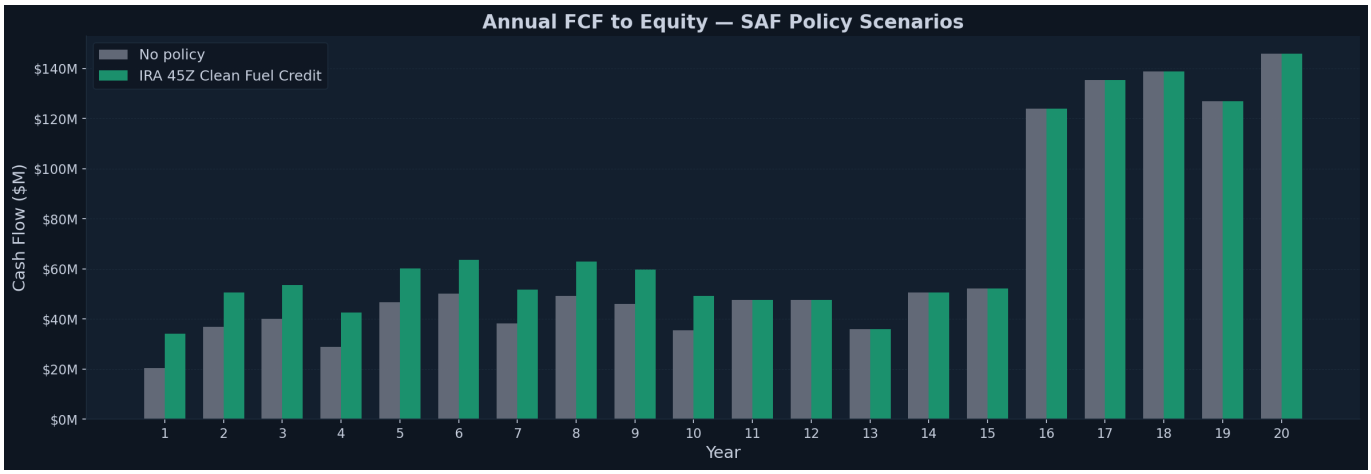
5. Policy Analysis

Policy Inputs (IRA 45Z)

Parameter	Value
SAF Sale Price	$6.0945/gal
Diesel Sale Price	$3.8990/gal
Naphtha Sale Price	$2.8391/gal
45Z SAF Credit	$0.350/GGE
45Z Non-SAF Credit	$0.200/GGE

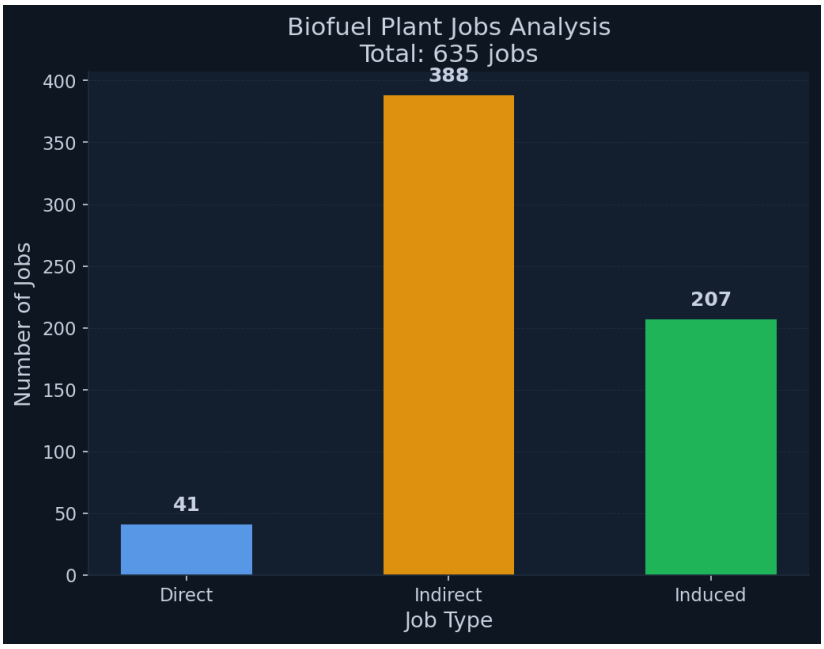
Credit Duration	10 yr
Jet-A Market Price	$3.104/gal
Diesel Market Price	$5.072/gal
Naphtha Market Price	$3.710/gal
Solve Required Credit	No

Metric	No Policy	With 45Z Credit
NPV (Equity)	$0.00M	$74.89M
Equity IRR	12.75%	15.69%
Payback	8.5 yr	6.5 yr
Revenue Yr1	$267.19M/yr	$267.19M/yr



6. Employment Impact

Category	Jobs	Description
Direct	41	Conversion facility operations
Indirect	388	Supply chain and services
Induced	207	Local economic spillover
TOTAL	**635**	


Biofuel Plant Jobs Analysis
Total: 635 jobs
Number of Jobs
400
350
300
250
200
150
100
50
0
41
388
207
Direct
Indirect
Induced
Job Type